EE5203 L08 Lab Guide

PWM — fading LEDs and controlling servos - Basri Erdogan

Mission: Breathe LD2 with TIM2 PWM, then position and sweep a hobby servo with TIM3—every `PSC/ARR/CCR` value chosen from the 16 MHz clock, every brightness change scheduled by the Week 7 tick, not by `HAL_Delay()`.

Prepare before class

- ☐ Re-read the theory slides up to "Positioning the servo."
- ☐ Bring your working L07 timer project (TIM2 tick interrupt).
- ☐ Know the answer to: with a 1 µs tick, what `ARR` gives 1 kHz? What gives 50 Hz?
- ☐ Servo group kit: SG90-class servo + 3 jumpers (5V, GND, signal).

Learning objectives

By checkoff time, you can:

1. configure a timer channel for PWM and justify `PSC/ARR/CCR` from 16 MHz;
2. show the alternate-function handover of a pin in the `.ioc` and the code;
3. change duty live—from code and from the SFR view;
4. command servo angles as pulse widths in microseconds, clamped to the contract.

Hardware and files

Item	Required value
Board	Nucleo-F401RE — LD2 **is** TIM2_CH1 (PA5): zero wiring for Part 1
LED PWM	TIM2 CH1, PSC `15`, ARR `999` → 1 kHz, µs units
Servo PWM	TIM3 CH1 on PA6 (D12), PSC `15`, ARR `19999` → 50 Hz
Servo wiring	red → 5V, brown/black → GND, signal → PA6 (D12)

Configure in CubeMX

Starting from a copy of the L07 project, saved as `L08_PWM`:

1. Click `PA5` → `TIM2_CH1`; **Timers → TIM2**: Channel 1 = `PWM Generation CH1`, PSC `15`, ARR `999`.
2. Click `PA6` → `TIM3_CH1`; **Timers → TIM3**: Channel 1 = `PWM Generation CH1`, PSC `15`, ARR `19999`.
3. Generate; find both AF assignments in the generated GPIO init before writing any code of your own.

Checkpoint A - A dimmed LED

```
HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_1);
__HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 350);   // 35 %
```

Evidence for Checkpoint A: LD2 visibly dimmer than full-on; in the SFR view, write `700` directly into `TIM2->CCR1` while paused and watch the brightness step—no code changed, no reflash.

Checkpoint B - The breathing LED

No `HAL_Delay()`: reuse the L07 tick. Every 10 ms flag from your timer interrupt, step the duty:

```
if (tick_10ms_flag) {
    tick_10ms_flag = 0;
    duty += dir * 5;
    if (duty >= 999) { duty = 999; dir = -1; }
    if (duty <= 0)   { duty = 0;   dir = +1; }
    __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, duty);
}
```

Evidence for Checkpoint B: smooth breathing; predict-then-check: what does changing the step from 5 to 20 do to the breathing period?

Checkpoint C - The servo obeys

At a 1 µs tick, `CCR` **is** the pulse width. Clamp to the contract:

```
uint16_t servo_us(uint8_t deg) {          // 0-180 deg -> 1000-2000 us
    if (deg > 180) deg = 180;
    return 1000 + (1000u * deg) / 180u;
```

```
}
HAL_TIM_PWM_Start(&htim3, TIM_CHANNEL_1);
__HAL_TIM_SET_COMPARE(&htim3, TIM_CHANNEL_1, servo_us(90));
```

Evidence for Checkpoint C: servo holds 0°, 90°, 180° on command; then a slow sweep stepped from the tick (not `HAL_Delay(20)` in a tight loop). Predict first: what `CCR` value is 45°?

Individual extension

Pick one, predict first:

- **Button-gated sweep:** the servo only moves while B1 (PC13) is held— combine with Week 6's EXTI or poll in the tick.
- **Gamma breathing:** linear duty steps look uneven to the eye; try stepping through a small squared lookup table and compare.
- **Second channel:** add TIM2 CH2 on PA1 with a different duty—two brightnesses, one timer, same frequency. Explain why the frequency *must* be shared.

Acceptance checklist

- ☐ `.ioc`: PA5 → TIM2_CH1 and PA6 → TIM3_CH1 (AF handover visible), PSC 15 both.
- ☐ LED at 1 kHz / servo at 50 Hz, both derived from 16 MHz on paper.
- ☐ Breathing scheduled by the timer tick — no `HAL_Delay()` in the loop.
- ☐ Servo commands clamped to 1000–2000 µs.
- ☐ One live duty change shown from the SFR view.

Individual oral check

- Why does the LED need `ARR = 999` but the servo `ARR = 19999`?
- What duty cycle is a 1.5 ms servo pulse—and why doesn't the servo care?
- Why can't the servo share TIM2 with the 1 kHz LED?
- `CCR = 1000` on the LED timer: what does the pin do? And `CCR = 0`?
- Where in the generated code did PA5 stop being a GPIO output?

Submit

Submit `main.c` and one SFR-view screenshot showing `TIM2->CCR1` mid-breath, plus your frequency derivations (16 MHz → PSC → tick → ARR) for both timers.

If you are stuck

Walk the chain: `PWM_Start` called for **that** channel? `CCR` actually nonzero? Pin really in AF mode in the `.ioc`? Then: **LED full-on/off only** — `CCR` outside 0…`ARR`, or you wrote to the wrong channel; **servo buzzes at the end stops** — commands outside 1000–2000 µs: clamp; **servo jitters** — power it from 5V (not 3V3) and share GND with the board; **breathing is jerky** — you are stepping from `HAL_Delay()` in a loop that also does other work; step from the tick flag instead.